Coarse-Graining to Investigate Cerebral Cortex Dynamics

The Case for Experimental Simplification of Biological Phenomena

Kentaroh Takagaki and Frank W. Ohl

Abstract Advances in multi-channel/multi-detector recordings and data analysis over the last decades have led to an explosion in the exploration of complex neural dynamics in mammalian cortex. Powerful methods have been applied to investigate such dynamics, including connectivity measures (correlation, causality, resting state synchrony, etc.), spatiotemporal pattern analyses, and finite-element modelling based on model neurons. These methods were initially applied to data from simple experimental models such as invertebrate neurons/ganglia/tecta, cell cultures, and organotypic slice preparations. Advances in the field have triggered the expanded use of such measures on more complex data, for example to mammalian ex vivo preparations, anesthetized preparations, and mammalian awake behaving preparations. With the increasing surgical, behavioral, and physiological complexity of the preparations themselves, less invasive measurement methods such as optical recordings, massively implanted arrays, or fMRI and other electromagnetic methods must be used to ensure robustness; however, these measures tend to feature lower signal-to-noise ratios, and are often prone to various biases. Furthermore, the high dimensionality of the data itself leads directly to potential errors in programming of analysis algorithms and overinterpretation of statistically significant but biologically insignificant findings. Given this situation, we advocate for the complementary use of the classical biological approach: the use of simplified preparations which may be limited in scope, but which highlight fundamental principles. We illustrate this approach with three experimental examples which use experimental and observational approaches to coarse-grain dynamic spatiotemporal activity patterns, to make coarse-graining observations of clinically relevant oscillations, and to coarse-grain complex behavior in mammalian discrimination learning.

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1 Putting the Mesoscopic Complexity of the Cerebral Cortex into Perspective: A Comparison with the Heart

In order to highlight the complexity problem we have described above and to illustrate the complexity of the cerebral cortex in particular, we start by comparing the physiological complexity of the cerebral cortex with that of the heart.

The heart is perhaps the best characterized mammalian organ in terms of its physiology and dynamics. Half a century of work has led to the characterization of the dynamics of, for example, basic pathological dynamics in atrial fibrillation [1, 2]. However, even in this extremely dramatic and common pathological dynamic, we are just now starting to unravel the causal influences of these atrial dynamics on the ventricle [3], to give an example. Even in the heart, we are far from being able to make quantitative predictions regarding the dynamic outcomes of even very basic pharmacological interventions, such as how an individual case of atrial fibrillation will respond dynamically to digitalis treatment.

Accepting that we have established that even the most basic of cardiac dynamics are just starting to be characterized, we can now illustrate the difficulty of handling cortical dynamics by comparing the cerebral cortex as a physiological system to the heart in terms of degrees of freedom.

We believe that it is justified to claim that the brain is astronomically more complex compared to the heart, in terms of its degrees of freedom (Table 1). One can start with the number of active elements, the basic unit of each dynamic system. It can be argued that the heart has only on the order of 10^0 active elements, since individual cardiomyocytes join into huge cardiac syncytia which are coupled both on the electrical and the cytoplasmic levels. Even if one counts individual cardiomyocyte nuclei as the basic unit, the number of nuclei in the heart is at least one order of magnitude less than the brain (Table 1). Next, one can think of the complexity of individual constituent units of each dynamic system. Whereas cardiac cells can be essentially classified into nodal cells and myocytes, even the most rudimentary classification of neurons will result in 10^1 transmitter classes and 10^2 anatomical morphological classes.

Spatially speaking, there is also a vast difference in degrees of freedom between the two dynamic systems. For example, the cardiac syncytium can essentially be handled as a warped two-dimensional excitable medium [2]. In contrast, the cortical mantle features not only a two-dimensional spatial distribution, but also a handful of topological degrees of freedom in the columnar (depth) dimension, as well as numerous proximal, short-range, and long-range projections, which can be considered as further dimensions in the cortical matrix, thus increasing the spatial dimensionality of the dynamic system. This difference in spatial dimensionality is clearly manifested by the types of clinical measurements utilized: for the heart, a 12-lead EKG will allow the description of most dynamic patterns, whereas even the coarsest measure of brain activity, the 10–20 EEG, features 19 leads, and this only allows unambiguous documentation of the simplest of dynamic patterns, such as epileptiform activity patterns and sleep slow waves.

Table 1 The brain is more complex than the heart. Various anatomical and functional measures are given to illuminate the magnitude of this difference

	Heart	Brain
Number of cells	10^0~ (nuclei: 10^{10})	10^{11}
Types of excitable cells	2	10^2–10^3
Spatial network dimensions	<2 spatial dimensions	>4 spatial dimensions
Connectivity	10^1/myocyte	10^3~/neuron
Temporal scales	1 temporal scale, fixed frequency for the most part	multiple irregular timescales spanning ≥ 6 orders of magnitude
Physiological oscillatory modes	Single	Desynchronized (δ, θ, α, β, γ, …)
Physiological outcomes of dynamic activity	Blood is pumped	Various sensations, motor planning, emotional states, hypothesizing, calculating, remembering, planning, daydreaming, empathizing, etc.
Dynamic pathological disturbances	Blood is not pumped effectively (atrial fibrillation, ventricular fibrillation, conduction block, ventricular tachycardia, atrial tachycardia/bradycardia, etc.)	Seizures (innumerable different types), loss of consciousness, amnesia, psychiatric disturbances (innumerable types), movement disorders, sleep disturbances, coma, migraine, etc.

Related to spatial dimensionality on the level of unit connectivity, the difference in degrees of freedom is also obvious: whereas cardiomyocytes are basically all interconnected as syncytia, cortical neurons form an interwoven network of mind-boggling complexity, as with the oft-cited number of 10^3 connections per neuron.

We can also illustrate the complexity of the cortex as a system based on the complexity of its functional outputs.

In terms of temporal dynamics, whereas the heart operates physiologically at a regulated rate of approximately 60–120 beats per minute, the cortical mantle operates not only at varying frequencies, but on at least 6 temporal scales: the sub-millisecond scale of STDP, the millisecond scale of spikes and spike rates, the second scale of metabolic changes as seen in fMRI and SPECT, the minute-to-hour scale of classical plasticity paradigms such as LTP and LTD, the daily scale of circadian activity patterns, and the days-to-decades scale of learning and forgetting as well as of cortical development and degeneration.

As physiologists, we cannot fail to mention the vast differences in the functional phenomenological complexity of the two dynamic systems. While the essence of cardiac function is limited to a unidirectional pumping of blood, the physiological essence of the cortical mantle encompasses a wild array of behavioral and neurological functions, spanning sensation in various modalities, planning and patterning of motor output for a vast array of both skeletal and smooth muscles throughout the

body, resting states such as sleep or posterior basic rhythm, as well as the myriad of cognitive and emotional states which make our lives so rich.

In terms of pathologies based on disruption patterns of dynamics, there is also a marked difference in the complexity of disruption patterns. Whereas the heart as an organ has only a handful of common failure modes, even a rudimentary list of common neurological and psychiatric disturbances would easily fill this page.

Taking all of the above comparisons together and summarizing them, it is a priori clear that the dynamics of the cerebral cortex are vastly more complex than that of the heart and have an astronomically higher degree of freedom, both on the constitutional substrate level and on the physiological phenomenological level. If one takes this together with the fact that even the vastly simpler cardiac dynamics have been explained to a limited extent, as argued above, it is no wonder that describing the neural dynamics of the cerebral cortex is a Sisyphean task.

Recent decades have seen the exponential growth of ever more precise measurement methods spanning from three-dimensional anatomical-physiological methods such as multiphoton imaging and MRI, to two-dimensional functional imaging methods such as single photon calcium and voltage-sensitive dye imaging, to metabolic methods such as intrinsic signal imaging, PET and fMRI, to electrical methods such as high-density EEG and massively parallel electrophysiology. We have seen a concurrent explosion in computational power, optimized algorithms (such as the vastly influential fast Fourier transform), and computational paradigms such as parallelization. Given such a large amount of information and information processing power, it is tempting to attempt bottom-up simulations of the brain from ground principles.

Valiant attempts have been made at bottom-up simulations of cortical dynamics from the unit level, most recently through the Blue Brain project [4]. However, the difficulty of the problem is illustrated by the result that even this most heroic of efforts by the most technically advanced computational neuroscientists has only succeeded in providing a partial reconstruction of a specific area of the cortex of a particular species, and only reproduces activity patterns resembling the most basic state transitions and cellular parameters. How does one then tackle the daunting task of understanding the dynamics of the brain? We argue that even in this age of massively parallel recordings and unprecedented information processing power, given the vast complexity of neural dynamics, that there is still persisting value in coarse-graining on a phenomenological level, i.e. to deal with biologically simplified systems.

Indeed, if one takes a step back, extrapolating from corner cases and simplified/special systems to understand general principles is a time-honored and vastly successful approach in biology: be it the study of genetics and reproduction in yeast or *E. coli* [5], the study of functional physiology in isolated organ preparations [6], or the study of evolutionary and adaptive principles in exotic ecosystems [7]. These approaches have provided insights which would not have been possible by directly studying human genetics, in vivo human physiology, or the full complexity of the Linnaean taxonomy, respectively.

For the field of neural dynamics, an equivalent biological approach is coarse-graining of observations. Indeed, this strategy has been fruitfully exploited,

and most of the ground-breaking studies in neurodynamics have been on simplified preparations. For example, while anesthetics had long been used in animal neurophysiology for practical reasons of animal handling during electrophysiological recording experiments, it was the notion that some anesthetics effectively function as "openers" for feedback loops in the sense of engineering control theory that has paved the way for entirely new approaches in experimental and theoretical neurophysiology [8]: With the experimental availability of techniques for opening and closing feedback loops in brain circuitry, far-reaching new conceptualizations of neuronal mass activity in paleocortex and neocortex became possible by applying concepts of (linear) control theory to (linearized) neuronal dynamics (for review, see [9, 10]). A similarly fundamental phenomenon of brain dynamics, the coordination of multiple rhythms, has been principally elucidated by the simple experiment of coordinating periodic finger movement [11]. It was possible to demonstrate with this approach that "synergetic" concepts, developed for describing self-organization of systems far from thermodynamic equilibrium [12], can be used to identify coordinated movements is this type of scenarios with order parameters emerging in a synergetic system of competing oscillators (for review see [13, 14]).

In this chapter, we advocate a renewed focus on biological and experimental coarse-graining, that is, a renewed focus on experimentally simplified preparations. We elaborate on the argument that advances in measurement and computation can best be taken advantage of by simultaneously coarse-graining our preparations experimentally, such that they manifest core biological phenomena in a simplified system, and provide robust experimental observations on a level amenable to in silico science. We illustrate this approach through three experimental systems which we have utilized in our recent work.

2 Example 1: Coarse-Graining of Emergent Dynamic Spatial Activity Patterns: Spiral Waves in the Epileptic Neocortex

A classical method of coarse graining in mesoscopic neurophysiology is to use preparations with simplified state dynamics, which are evoked using pharmacological agents. This can be either with explicit pharmacological blocking of certain receptors, or implicit global state changes with anesthetic agents.

Pharmacological studies of cortical dynamics have traditionally focused on agents which create clear oscillatory or epileptiform states, such as interictal-like spiking elicited by the $GABA_A$ inhibitor bicuculline. Although the dynamics of such states tend to be vastly simplified, the clear spiking/oscillatory nature of the preparation is conducive to the analysis of spatial connectivity and propagation (e.g. [15]). One can also block certain pharmacological subcircuits of the cortex in vivo to try to elucidate the roles of different subcomponents (e.g. [16]). Of note, given the complex dynamics of the cortex, even the most delicate pharmacological

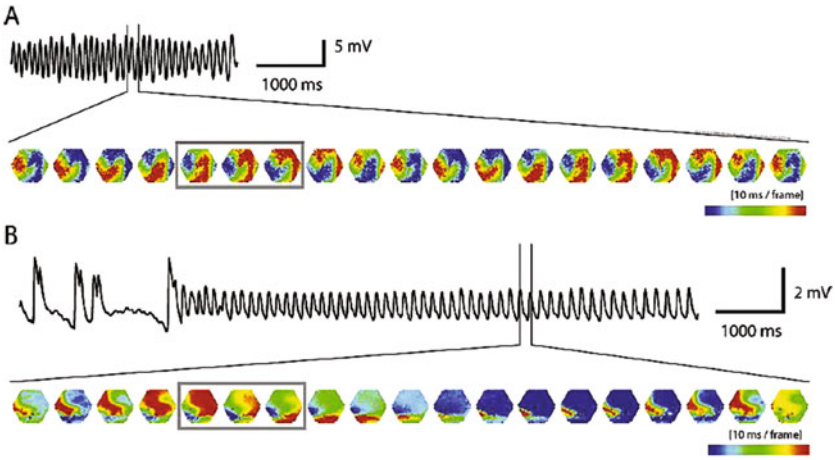

Fig. 1 Spiral activity in the neocortex as seen with voltage-sensitive dye imaging. Two examples of stable spiral dynamics are demonstrated, first in an organotypic slice preparation treated with carbachol and bicuculline (**a**), and next in an epileptiform intact cortex from an epileptic Mongolian gerbil, recorded through a craniotomy. From [19], with permission from Elsevier

blockades of the cortical network tend to lead to massively simplified and coarse-grained dynamic patterns, amenable to reductionistic analysis.

One recent line of work using a weakly coupled oscillatory state induced by bicuculline and the cholinergic agonist carbachol, for example, has shown that spiral dynamics can serve as an organizing principle for mesoscopic activity within the cortical mantle [17]. This is an example where experimental coarse graining of cortical dynamics has illustrated a new dynamic principle in cortical organization. Although one may question the extent to which such organizing principles are relevant for non-pharmacologically treated intact cortex, we are guardedly optimistic about the general significance of our findings given that the phenomenon, which was first discovered in organotypic slice preparations, has now been seen in in vivo pharmacologically treated preparations as well as in sleep-like states of light anesthesia [18]. We have also observed such activity in epileptiform cortical activity patterns of Mongolian gerbils with epilepsy (Fig. 1b; [19]). Since the spiral dynamics are much less stable in vivo than in drug-treated slice preparations, one can very well imagine that without initial coarse-grained study in non-physiological organotypic slice preparations, the phenomenon had very little chance of being observed at all.

Anesthetic states provide a somewhat more physiological means to curtail the dynamic degrees of freedom inherent in the cortex. Given the "burstiness" of anesthetized cortex in general (true anesthetic agents invariably lead to cortical depression and rebound bursting), these simplified preparations allow study of the transmission of large bursting impulses of population activity (e.g. [20]).

By modulating the depth of anesthesia, one can attain dynamics which are somewhat less simplified, yet still stable. For example, slow withdrawal of pentobarbital anesthesia leads to sleep-like cortical states with theta and delta wave domination; we have recently shown that such states can sustain short in vivo bouts of spiral dynamics [18], which we speculate may play a role to stabilize recurrent cortical dynamics during memory consolidation in sleep. Modulating the depth of urethane anesthesia leads to a biphasic state of slow-wave deep sleep-like activity and fast desynchronized REM sleep-like activity, and one can take advantage of such simplified dynamics to study the lateral propagation of mesoscopic activity [21]. Based on such findings, one can speculate as to the extrapolability to analogous physiological states of deep sleep and REM/wakefulness.

3 Example 2: Coarse-Graining of Human Mesoscopic Brain Dynamics: Posterior Basic Rhythm in the Development of Autism

Pharmacological simplifications such as in the previous section are rarely possible for studying the human brain, where pharmacological interventions are reserved for medical indications. Even if a medically necessary pharmacological treatment fortuitously simplifies cortical dynamics, such as say with pregabalin-induced increase in slow-wave sleep [22], various uncontrollable factors such as co-medication and disease state will necessarily complicate interpretation of results.

Although it is increasingly fashionable to measure complex metrics of cortical dynamics such as resting state connectivity, for example, these measures feature a low signal-to-noise ratio, and are much more susceptible to potential multiple testing and other statistical errors. Given the complexity of these measures, we believe that there is enduring value in focusing instead on the simplest of human brain dynamics and conducting coarse-grained observations at this level.

One classical method of coarse-graining human cortical dynamics is to drive the system with entraining stimuli. Stimuli driving strong dynamics are usually regular, such as photic stimulation eliciting epileptiform oscillatory bursting. This is used regularly in clinical practice to provoke and document non-overt epileptiform dynamics in patients with epileptic habitus [23]. Another classical experimental approach is to use continuous tone sequences which drive regular population activity in auditory areas of the brain; by occasionally changing these driving stimuli, for example by dropping stimuli or inserting “oddball” stimuli within the driving train stimulus, one can investigate excursions from a mesoscopically entrained and stable brain dynamic [24].

Another approach is to use highly synchronized brain states to the advantage of investigative questions. To this end, we have recently focused on one of the most classic and one of the most robust of human brain dynamics, the posterior basic EEG rhythm of Berger [25]. This very stable dynamic emerges in the posterior pole

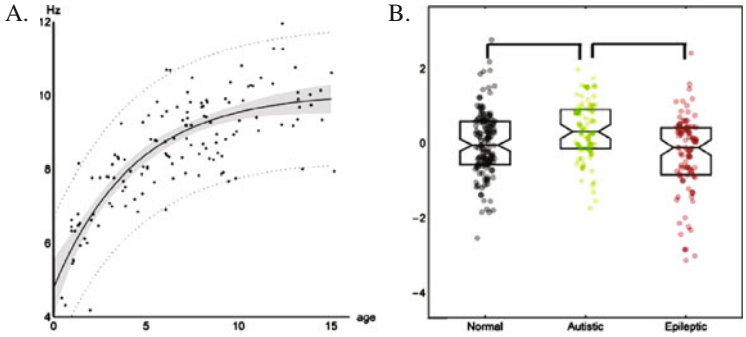

Fig. 2 The development of the posterior basic rhythm (PBR) is accelerated in autistic children. **a** Development of PBR in normal children (n = 134) who were referred to our center for evaluation of inattentiveness and other issues, which turned out to have a non-psychiatric diagnosis. **b** Corrected PBR frequency in developing children shows statistically significant increase in autistic children (*green*), and statistically significant decrease (developmental delay) in epileptic children (*red*). We believe that these statistically significant results may also be biologically and clinically significant. From [26], with permission from Elsevier

at around several months of age at a frequency of 3–4 Hz, and gradually develops over the course of 8–10 years of life to reach the adult "alpha" range of 8–12 Hz. Functionally, this stable dynamic is characterized by high amplitude in a quiet restful state with eyes closed, and is attenuated by eye opening. We focused on this most simple of cortical dynamics recorded by EEG, and retrospectively analyzed this state in children with autism and normal controls (We also studied epileptic children, given the high incidence of overt and covert epileptiform activity reported in autistic children.) By quantifying even the most simple characteristics of this state such as base oscillation frequency, we concluded that children with autism (on a retrospective cohort base) show faster development of the posterior basic rhythm than their non-autistic peers (Fig. 2).

4 Example 3: Coarse-Graining of Complex Behavior

Another dimension of complexity which must be dealt with in neuroscience is the diversity of behavior. For example, even the simplest tasks in experimental human psychology feature a large degree of hidden complexity, in that the subject is verbally appraised of the task context (e.g. sitting in a chair in a quiet room), the modality in question (e.g. looking at the monitor), as well as the general timescale of the experiment (e.g. "you will be asked to answer questions for about 30 min"). Such covert contextual cues cannot be given to animals, much less advanced linguistic descriptions of task contingencies.

A further dimension of complexity is the unitary nature of experience. Even for relatively simple tasks, it has been shown that operant responses are dependent

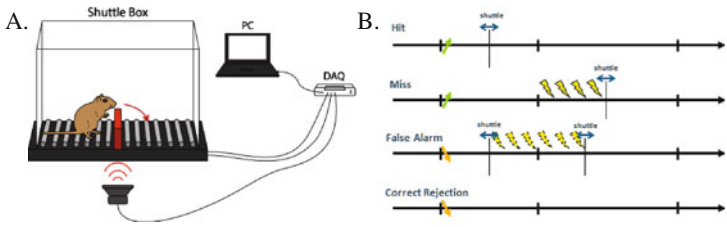

Fig. 3 The Mongolian gerbil auditory Go/NoGo shuttlebox task. **a** The animal performs the task in a two-chamber shuttlebox, with hurdle. We use a hurdle here (in contrast to the tunnels more common in shuttleboxes for mice), to take advantage of the species-specific tendency of gerbils to jump. The grid floor delivers a mildly aversive footshock to provide negative reinforcement to the animal. **b** Timeline of four possible behavioral outcomes. For Go trials, the animal will receive a certain tone, in this case a rising FM sweep (*light green*). If the animal shuttles to the other compartment within a fixed period of time, this is a correct "hit" trial (*first row*). If the animal fails to shuttle in response to a Go tone, this is considered a "miss" trial, and it will receive footshock until it escapes. For a NoGo tone (in this case a falling FM sweep; *orange*), the animal must remain in its current shuttlebox compartment (*fourth row*; "correct rejection"). If it shuttles, it will receive a mild footshock (*third row*; "false alarm")

upon the proximal history of reward [27]. Such history effects are further complicated by learning paradigms, where the animal's behavior changes progressively. Sometimes, sudden changes in behavior can mark the animals' mastery of the operant contingency, what we refer to the "aha" moment [28].

How can one simplify complex behaviors, and coarse grain them such that their dynamics are amenable to analysis? We have approached simplification of rodent auditory discrimination/categorization by using a Go/NoGo shuttlebox paradigm (Fig. 3). Using this task, we are able to simplify behavioral response contingencies into the four categories of signal detection theory, namely, hit, miss, false alarm, and correct rejection. The key point in terms of behavioral coarse-graining here is that the discrimination is manifested in a purely binary decision: whether to jump (Go) or to stay (NoGo). This binary nature reduces the complexity of the task when compared to more common rodent sensory two-choice discrimination tasks such as T maze or two nosepoke choice tasks, as these tasks are not strictly binary; they actually constitute a trinary choice task: go left, go right, or do not go, which complicates the behavioral readout in terms of signal detection theory.

By coarse-graining the behavioral paradigm in a manner tailored to our question (i.e. by simplifying our task to mirror our theoretical framework of signal detection theory), we were able to demonstrate spatiotemporal patterns in cortical ECoGs which predict, in a manner not time-locked to sensory stimulus, the animal´s behavioral contingency in single trials based on the framework of signal detection theory [28].

5 Summary

In this chapter, we have argued that experimental coarse-graining is still necessary to effectively tackle the vast complexity presented by the cerebral cortex, even with modern advances in computational capacity and algorithmic efficiency. We have supported our arguments by comparing cortical dynamics to the vastly simpler yet nonetheless highly complex dynamics of the heart. We illustrate our approach by describing three experimental systems which have provided us insights into cortical dynamics which would not have been possible through the more fashionable approach of attempting bottom-up induction from single-neuron dynamics or by using complex methods to analyze fine-grained data.

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